ASSIGNMENT 7 · KAGGLE COMPETITION

Predicting Heart Disease

Tree-based models on 34,300 patients

Carter Deslauriers

Missing data: fill it, and flag it

Column	Missing	%
family_history	2,028	5.9%
sleep_hours	1,747	5.1%
waist_cm	1,635	4.8%
resting_hr	1,313	3.8%
weight_lb	1,044	3.0%
cholesterol_level	688	2.0%

Why not drop rows?

7,619 rows (22%) have a blank somewhere. Dropping them throws away a fifth of the data.

What we did

Fill each blank with the column median, and add a 0/1 “was missing” column so the model can tell.

Recording errors: fix what we can

Problem	Example	What we did
Negative values	−135	Dropped the minus sign → 135
Extra digits	14010, 995	Dropped digits → 140, 99
Diastolic > systolic	80 / 120 (603 rows)	Swapped them
Still unrealistic	Height 24 in	Set to missing, filled like the rest

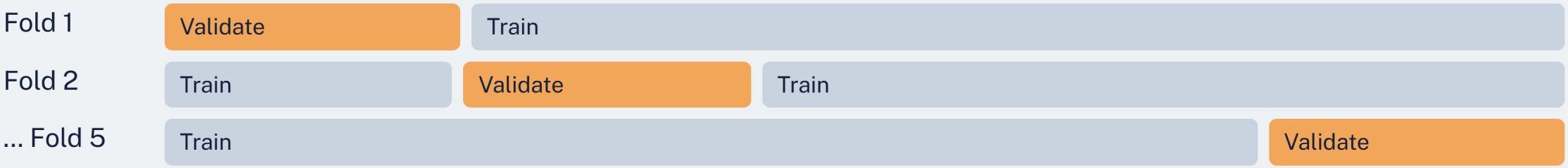
Realistic ranges: systolic 80–220, diastolic 40–140, height 48–84 in, weight ≥ 80 lb. **Only about 1% of blood pressure readings had to become missing.**

Models we tried

Model	Why we tried it
Decision tree	Simple, explainable baseline
Random forest	Many trees vote; averaging reduces variance
Extra trees	Random split points add more variety between trees
LightGBM	Boosting: each tree fixes the last one's mistakes
XGBoost	Most-cited boosting library on tabular benchmarks

An unlimited-depth tree scored only **61.7%**: it memorized the training patients (overfitting).

How we split the data: 5-fold cross-validation



- Only train.csv; each fold is 80% train / 20% validate
- Stratified: every fold keeps the 50/50 yes/no balance
- Metric: accuracy (the Kaggle metric)
- **No leakage:** medians are computed inside each fold from the training part only

Tuning: tree, forest, extra trees

Grid search with 5-fold CV · every combination tried

Model	Hyperparameter	Values searched	Best
Decision tree	max_depth	3, 5, 7, 10, None	7
20 combos · 0.7148	min_samples_leaf	1, 20, 50, 100	100
Random forest	n_estimators	200, 400	400
18 combos · 0.7255	max_depth	8, 12, None	12
	min_samples_leaf	1, 10, 30	30
Extra trees	n_estimators	200, 400	400
18 combos · 0.7186	max_depth	8, 12, None	None
	min_samples_leaf	1, 10, 30	10

Tuning: gradient boosting

Grid search with 5-fold CV · 88 combinations in total across all five models

Model	Hyperparameter	Values searched	Best
LightGBM	n_estimators	200, 500	200
16 combos · 0.7257	learning_rate	0.03, 0.1	0.03
	num_leaves	15, 31	15
	min_child_samples	20, 100	100
XGBoost	n_estimators	200, 500	500
16 combos · 0.7265	learning_rate	0.03, 0.1	0.03
	max_depth	3, 5	3
	subsample	0.8, 1.0	0.8

Final model: XGBoost

Validation (5-fold CV)

72.65%

Public leaderboard · 4th place

73.17%

- **Why:** highest CV score, and boosting is the proven best on tabular data
- **Honest caveat:** XGBoost, LightGBM and random forest are within 0.001, which is noise
- **Trained on:** all of train.csv, after tuning was finished